JOURNAL OF MEDICAL MICROBIOLOGY

RESEARCH ARTICLE

Sawant *et al.*, *Journal of Medical Microbiology* 2021;70:001417
DOI 10.1099/jmm.0.001417

MICROBIOLOGY SOCIETY

Identification & correlation of bacterial diversity in oral cancer and long-term tobacco chewers- A case-control pilot study

Shriya Sawant[1], Jinesh Dugad[2], Deepak Parikh[2], Sathiyaraj Srinivasan[3,4,*] and Harinder Singh[1,*]

Abstract

Introduction. Squamous cell carcinoma is a highly aggressive type of oral cancer (OC). It is the most common cancer among men, and accounts for almost 90% of all oral cancers in India. Consumption of tobacco is a leading factor contributing to maximum oral cancer incidences as per the WHO.

Hypothesis/Gap statement. Researchers reported a direct association of microorganisms with dysbiosis in various oral lesions including oral cancer. However, there is a dearth of information related to compositional changes in the oral microbiome in long-term tobacco chewers and the Indian oral cancer population.

Aim. The aim of this study was to identify and correlate the bacterial diversity in the oral cavity of tobacco chewers, patients with oral cancer and healthy subjects in the Indian population.

Methods. Oral rinse samples were collected for ten subjects in each group followed by DNA extraction. The variable regions of the bacterial 16S rRNA gene (V6-V8) were amplified, sequenced, processed, and analysed using QIIME2 platform to assess alpha and beta diversity between the study groups.

Results. This pilot study showed genus *Streptococcus* dominated the control group (18.54%), and the abundance decreased in tobacco and OC group (9.63 and 5.45% respectively); whereas genus *Prevotella* dominated the tobacco and OC group (21.01 and 26.03% respectively). A shift in abundance of microbiome was observed from control population to oral cancer via the tobacco chewing population. Maximum alpha diversity of oral microbiome was found in Indian tobacco chewers. Beta diversity of tobacco chewers was similar to both the healthy population as well as oral cancer patients suggesting transitioning of the oral microbiome from healthy to oral cancer microbiome via the tobacco chewers microbiome.

Conclusion. The data provides evidence of oral bacterial dysbiosis due to tobacco chewing habits that can further lead to progression towards cancer.

INTRODUCTION

The human body is a complex ecosystem of human cells and microorganisms. A huge diversity of microbial populations are found in the human body and called the microbiota. This microbiota provides healthy associations and supports good health [1]. The oral cavity, with its various hard and soft anatomic surfaces, makes the oral environment conducive for the growth of the different types of microbes and harbours more than 700 species of bacteria [2]. But, continuous exposure to external agents, a sedentary lifestyle and dietary habits can affect the human microbiota in terms of its diverse profile, causing dysbiosis which refers to the alterations in the bacterial composition [3, 4]. Dysbiosis in the oral cavity has been linked to various diseases, including cancers [5–8].

The risk factors for oral cancer include consumption of tobacco and tobacco products, smoking, alcohol consumption, poor oral hygiene and viral infections like HPV (Human papilloma virus) [9, 10]. In South Asia, 250 million people use

Received 15 February 2021; Accepted 28 July 2021; Published 23 September 2021
Author affiliations: [1]Department of Biological Sciences, Sunandan Divatia School of Science, NMIMS University, Vile Parle (West), Mumbai, India; [2]Somaiya Ayurvihar -Asian Cancer Institute, Off Eastern Express Highway, Behind Everard Nagar, Somaiya Ayurvihar, Sion (East), Mumbai, India; [3]Department of Bio & Environmental Technology, College of Natural Science, Seoul Women's University, Seoul 01797, Republic of Korea; [4]Gene Strand Technologies Pvt. Ltd., Chennai, India.
***Correspondence:** Sathiyaraj Srinivasan, drsrini@swu.ac.kr; Harinder Singh, Harinder.Singh@nmims.edu
Keywords: Dysbiosis; Microbiome; Oral cancer; Tobacco.
Abbreviations: OC, oral cancer; OTU, operational taxonomic unit; QIIME, quantitative insights into microbial ecology.
One supplementary table is available with the online version of this article.

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smokeless tobacco on a regular basis and tobacco chewing has been linked to increased risk of oral cancer by a factor of six, thereby making it the most common risk factor for oral cancer in Asia [11, 12]. Apart from these, 15% of oral cancer cases cannot be linked to the above-mentioned risk factors, which has increased the need to detect more risk factors [13]. There is an increasing evidence of the role of bacteria in the metabolism of carcinogens, such as tobacco-specific-N-nitrosamines which emerge from tobacco products [14].

Oral cancer continues to be one of the leading cancers with an ever-increasing incidence as well as mortality rate. According to the Globocan 2018 data, India contributes to about 33.81% of global cases and is the second most common cancer in the country. Oral Squamous Cell Carcinoma (OSCC) makes up for 90% of oral cancer cases that are the most commonly found in Indian males (https://gco.iarc.fr/today/home). With the steep rise in the oral cancer cases in the Indian subcontinent, and the associated involvement of tobacco usage, it is necessary to evaluate the role of the oral microbiome in these cases [9, 15, 16].

Along with these factors, the role of the oral microbiome in oral cancer has been unveiled by the Human Microbiome Project (HMP) [17, 18]. With the help of next-generation sequencing (NGS) technology, the identification and quantification of the microbiome including bacteria, viruses, and fungi in healthy or diseased individuals, have become effortless [19, 20]. Indeed, numerous metagenomics studies of the microbiome have highlighted microbial pattern modifications in various types of cancer and viral infections. Reports have also been published depicting diversity in the oral microbial flora depending on the stage of oral cancer [21], the malignant lesions and disorders of the oral cavity [22], and mutational changes in oral cancer [23]. However, there is a dearth of genomic study on the bacterial community composition in the oral cavity and its comprehensive profile in cancer in the Indian population.

The purpose of this study is to analyse the typical bacterial composition and diversity of the oral microbiome in patients with oral cancer and long-term tobacco chewers in the Indian population, using the next generation sequencing (NGS) technology. We hypothesized that the oral microbiota community could dynamically change in OSCC patients. To the best of our knowledge, no previous reports have been published describing the oral microbiome associated with oral cancer patients and the tobacco chewers in India.

METHODS

Subject recruitment and data collection

Thirty subjects in the study were classified into three groups - Healthy (control, *n*=10), OSCC (OC, *n*=10) and long-term tobacco chewers (tobacco, *n*=10), each consisting of ten individuals. All the participants enrolled in the study were 18 years or older. The healthy controls were defined as non-smokers and without any diagnosed diseases in the oral cavity, which was confirmed by prior clinical examination. Long-term tobacco chewers were individuals chewing various forms of tobacco for 5 years or more and without any oral or periodontal disease. All diagnoses for patients with OSCC were confirmed by biopsy and pathological findings. All control and long-term tobacco chewing participants were devoid of any antibiotic treatment at the time of sample collection. The samples were collected in the period of January 2018 to March 2020.

Data on participant's medical history, age, gender, occupation, habits of tobacco chewing, alcohol intake, and general oral hygiene questions were recorded. Informed written consent was obtained prior to the sample collection from all the participants. Ethical approvals for the recruitment of healthy, long-term tobacco chewing volunteers and oral cancer patients were granted by SVKM's Institutional Ethics Committee (NMIMS/IEC/008/2016) and The Ethics Committee of K. J. Somaiya Medical College and Hospital, Mumbai.

Sample collection

Patients rinsed their mouths with 20 ml of sterile normal saline for 30 s and spit into a 50 ml sterile tube. Salivary samples from patients were taken at the pre-operative stage while the patient was NBM (Nill By Mouth). Subjects from healthy and tobacco groups were asked to refrain from eating, drinking, or oral hygiene procedures for at least 1 h before sample collection. Salivary samples were collected in well-labelled sterile falcon tubes and stored at 4 °C and processed within 48 h.

DNA Extraction and quantitation

The collected samples were centrifuged at 14000 r.p.m. for 8 min at room temperature. The supernatant was discarded and the pellet was used for DNA isolation by using Invitrogen PureLink Genomic DNA kit (Cat no. K182002) according to the manufacturer's recommendation. DNA was eluted using 100 µl elution buffer and was stored at −20 °C until PCR amplification. DNA sample integrity was assessed by electrophoresis on 1% agarose gel. The purity and yield of DNA were assessed spectrophotometrically using BioTek EPOCH2 microplate reader. The purity of DNA was determined by the ratio of absorbance at 260 nm and 280 nm (A260/A280).

Next generation sequencing

The PCR amplification of bacterial 16S rRNA hypervariable region V6-V8 was carried out using primers B969F (ACG CGH NRA ACC TTA CC) and BA1406R (ACG GGC RGT GWG TRC AA) [24, 25]. The V6-V8 region has been accepted as a low error-prone region for taxonomic assignment and community clustering [26]. The PCR reaction was carried out by 30 s initial denaturation at 98 °C, 30 cycles of 10 s denaturation at 98 °C, 30 s annealing at 55 °C, 30 s elongation at 72 °C, and a 5 min final extension at 72 °C. The whole sequencing process was done using Illumina (Illumina, San Diego, CA). The library was prepared by standard library construction protocol (https://support.illumina.com/downloads/

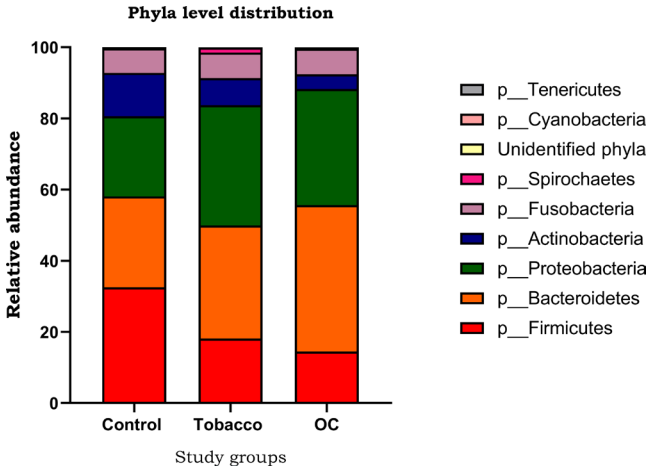


Fig. 1. Phyla-level distribution of bacteria within groups.

16s_metagenomic_sequencing_library_preparation.html) using Nextera XT kit (Illumina, San Diego, CA), following the manufacturer's instructions. The specific amplicons for V6-V8 region were quantified in each reaction mixture along with Illumina sequence adapter, and index primers (Nextera XT Index kit) were utilized in emulsion PCR to generate amplicon libraries followed by PCR clean up. MiSeq libraries were quantified and then subjected to 300-nucleotide paired-end multiplex sequencing on Illumina MiSeq.

Bioinformatic analysis

Demultiplexed raw reads (V6-V8) from Illumina sequencer were obtained and imported into Quantitative Insights into Microbial Ecology 2 (QIIME2) pipeline for further analysis. The quality of the reads from the sequencer were assessed using FASTQC. The reads for each sample were paired using VSEARCH and the quality was assessed again using FastQC. Low quality reads (Phred score Q<30) were filtered out followed by denoising and assignment into Operational Taxonomic Units (OTUs) by Deblur. Taxonomy was assigned using feature-classifier-sklearn against the Greengenes database (gg_13_5, https://greengenes.secondgenome.com/?prefix=downloads/greengenes_database/gg_13_5/) at different taxonomic levels. Bacterial richness and diversity were analysed by alpha and beta diversity matrices and indices such as Observed OTUs, Shannon index, Chao index, ACE index, Rarefaction curves, Weighted and Unweighted Unifrac distance matrices, and PCoA plots.

Observed OTUs exist based on sequence data, i.e. similar sequences are clustered into a single OTU. Based on OTUs, Shannon and Simpson diversity indices are measured to estimate bacterial diversity in a sample and provide more information about the bacterial composition as compared to simple species richness or evenness. On the other hand, Rarefaction, ACE and Chao1 indices estimate richness of a sample [27]. Pielou_e indicates an evenness measure which describes how evenly bacterial communities are distributed in the samples and Good's coverage index depicts sufficient sampling for the data [28, 29].

Statistical analysis

The differential abundance of bacterial genera was analysed using LEfSe (https://huttenhower.sph.harvard.edu/galaxy/). The Lda Effect Size was analysed using Kruskal Wallis test (threshold=0.05) and Pairwise Wilcoxon test (threshold=0.05). To analyse the beta diversity, PERMANOVA, ANOSIM and PERMDISP was performed using QIIME2 pipeline. Graphpad Prism 8 was used to carry out statistical analysis of the results obtained.

RESULTS

Clinical data

Ten volunteers from each group (healthy volunteers, long-term tobacco chewers, and patients with oral cancer) were recruited in the study. Of these ten, three samples, one from each of the study groups failed the sequencing procedure and therefore their data is not included in the results below. The demographic data of the volunteers is tabulated in Table S1 (available in the online version of this article).

Microbial richness and diversity

In the study V6-V8 hypervariable regions of 16S rRNA were sequenced using the Illumina MiSeq system. A total of 1444088 raw reads of 300 bp length were generated. After joining the read pairs and filtering out low-quality sequences (Phred score Q<30), 1231511 high-quality sequences were retained. After joining the forward and reverse reads, sequences of 430 bp length were obtained which were then trimmed to 300 bp length due to drop in quality of the sequences beyond 300 bp with an average number of reads 45611 reads per sample which were classified into 1719 OTUs using Greengenes database (version gg_13_5).

After assigning taxonomy, nine phyla, 15 classes, 22 orders, 44 families, 66 genera, and 101 species were identified. The top five phyla observed were Firmicutes, Proteobacteria, Bacteroidetes, Fusobacteria, and Actinobacteria. Maximum abundance of phyla Firmicutes was observed in the control group (32.56%), phyla Proteobacteria in tobacco (33.86%), and phyla Bacteroidetes in the OC group (41.13%) (Fig. 1). The ten most abundant genera observed in all groups were *Streptococcus, Prevotella, Rothia, Neisseria, Haemophilus, Porphyromonas, Veillonela, Leptotrichia, Fusobacterium,* and *Granulicatella*. Of these, genus *Streptococcus* was most abundantly present in control samples (18.544%), whereas the abundance was decreased in the tobacco chewing group and OC group to 9.634 and 5.455% respectively (Fig. 2). Genus *Prevotella* dominated the tobacco (21.009%) and OC group (26.034%). Genus *Porphyromonas,* which is a known major oral periodontal pathogen was found in increased abundance in OC patients (11.621%) as opposed to tobacco chewers (7.696%) and control population (5.610%). The relative abundance of the ten most abundant genera is shown in Fig. 3. The other oral pathogens such as those belonging to genus

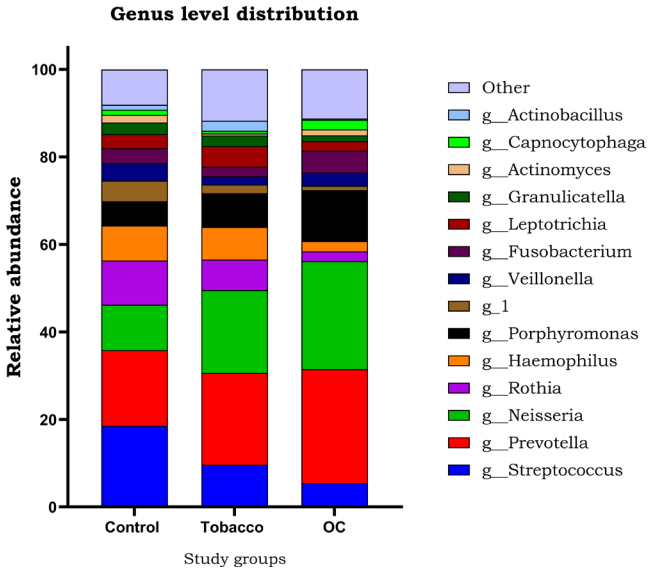


Fig. 2. Genus-level distribution of bacteria within groups. 'Other' contains the following genera grouped together: *Aggregatibacter, Paludibacter, Lautropia, Camplylobacter, Tannerella, Lactobacillus, Gemella, Treponema, Dialsiter, Atopobium, Oribacterium, Parvimonas, Corynebacterium, Filifactor, Peptostreptococcus, Abiotrophia, Megasphaera, Bulleidia, Selenomonas, Halomonas, Eikenella, Scardovia, Bifidobacterium, Moryella, Shuttleworthia, Catonella, Mycoplasma, Lachnoanaerobaculum, Pseudomonas, Propionibacterium, Bacteroidetes, Morganella, Alloscardovia, Peptococcus, Klebsiella,* g_2, g_3, g_4, g_5, g_6, g_7, g_8, g_9, g_10, g_11, g_12, g_13, g_14, g_15,g_16, g_17, g_18,

Aggregatibacter, Prevotella, Parvimonas, Campylobacter, and *Peptostreptococcus* were also found in increased abundance in OC patients or tobacco chewers as compared to control population.

On comparing the diversity at the genus level, few genera were found to be explicitly present in one or two of the three groups (Fig. 4). Genera *Abiotrophia* along with four unidentified genera were present in control and tobacco group; *Shuttleworthia* and *Bifidobacterium* were present in samples of control and OC group, whereas genera *Catonella, Lachnoanaerobaculum* and *Mycoplasma* was exclusively present in OC and tobacco group. Genera *Pseudomonas*, *Peptococcus, Morganella, Alloscardovia, Klebsiella, Bacteroides,* and *Propionibacterium* were present in samples from oral cancer patients only. Since species level taxonomy was not well attained, the data has not been included.

Alpha and beta diversity comparisons

To understand the species richness and diversity of samples, alpha (within-sample) diversity matrices were calculated using observed OTUs, Ace index, Chao1, Goods coverage, Shannon and Simpson indices, and Pielou_e. The Good's coverage for all the groups was greater than 98%. Rarefaction curves displayed plateau for species richness (up to 350 OTUs) was reached at approximately 6000 reads per sample. The alpha diversity was slightly higher in the tobacco chewing group as compared to the control group, although this result was non-significant. In contrast, alpha diversity is comparatively lower in the oral cancer group as compared to both the control group and the tobacco chewing group. As compared to the tobacco group, the OC group had lower alpha diversity based on Observed OTUs (mean number of OTUs 277 and 180 respectively, P=0.0071), ACE index (P=0.0022) as well as Shannon index (P=0.0326) (Fig. 5) (Table 1).

Beta diversity (between-group) was studied using various parameters as mentioned in Table 2 and shows significant differences between the three groups. Unweighted Unifrac distances take into account the qualitative differences/diversity between the groups of study whereas Weighted Unifrac distances account for the diversity between groups quantitatively. Between control and OC groups, unweighted unifrac distances significantly differed based on PERMANOVA (P=0.017), ANOSIM (P=0.011), and PERMDISP (P=0.016). Significant differences in unweighted unifrac were also observed between control and tobacco groups based on PERMANOVA (P=0.009) and ANOSIM (P=0.018) along with significant differences between OC and tobacco group based on PERMANOVA (P=0.003), ANOSIM (P=0.008) and PERMDISP (P=0.012).

PCoA plots show significantly differential clustering of OC samples as compared to tobacco and control samples which have clustered together (unweighted unifrac-qualitative). Similarly, this kind of clustering together of control and tobacco samples is observed in weighted unifrac (quantitative) PCoA plots, Bray-Curtis matrix (quantitative) as well as Jaccard matrix (qualitative) (Fig. 6). The unifrac PCoA plots take into account the phylogenetic distances between the OTUs present in the sample, whereas Bray-Curtis and Jaccard matrices take into account the community dissimilarity in the samples, without looking at the phylogenetic distances between them. The presence of similar bacterial species in control and tobacco groups as compared to the OC group can be inferred from the PCoA plots, thereby suggesting the transformation of bacterial diversity from control to tobacco to the OC group. This observation positively correlates and supports the fact that tobacco chewing is a very important risk factor for oral cancer. No significant differences were found in the weighted unifrac based on PERMANOVA, ANOSIM, and PERMDISP on comparing all three study groups to each other.

LEfSe analysis carried using Kruskal-Wallis test and Wilcoxon test displayed no differentially abundant taxa which could act as a biomarker for this study.

DISCUSSION

Numerous reports have been published displaying the diversity and variation in oral microflora in various conditions including oral cancer. Most of these studies have reported an increase in the abundance of pathogenic bacteria in the oral cavity of oral cancer patients as compared to healthy individuals. The oral microbiome of tobacco chewers and its

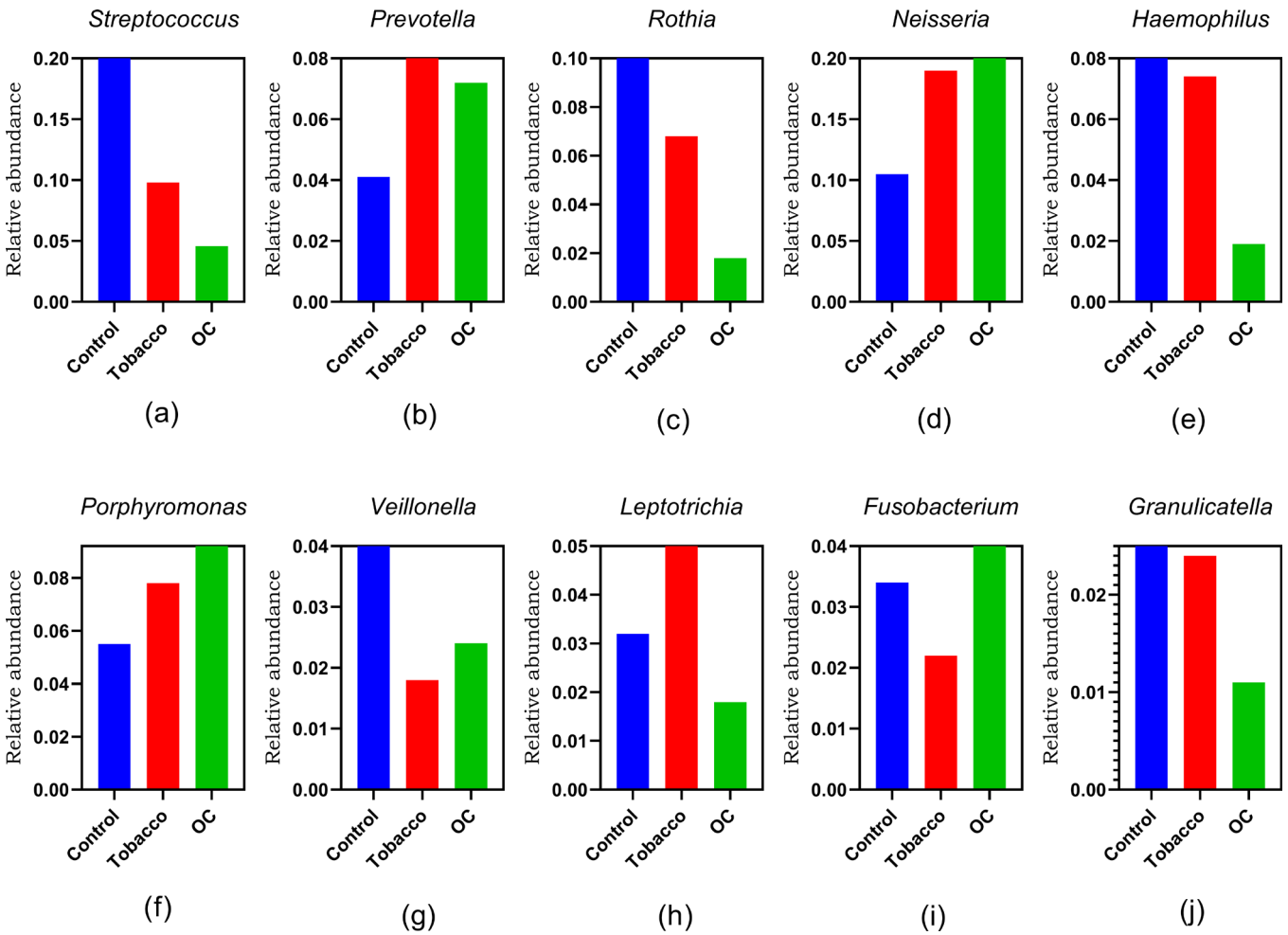
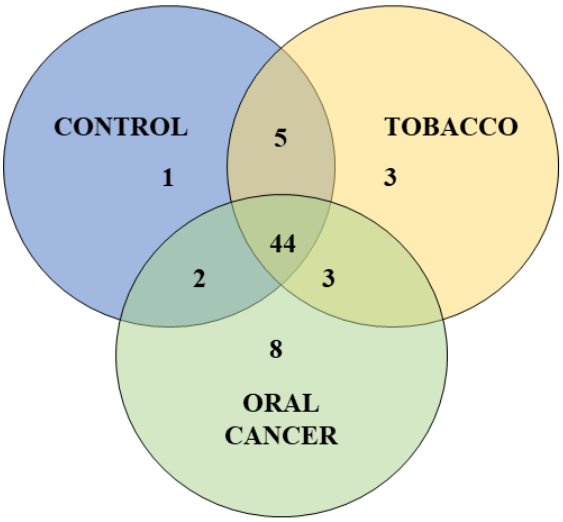


Fig. 3. Relative abundance of ten most abundant genera found in the study between the three study groups; (a) *Streptococcus*, (b) *Prevotella*, (c) *Rothia*, (d) *Neisseria*, (e) *Haemophilus*, (f) *Porphyromonas*, (g) *Veillonella*, (h) *Leptotrichia*, (i) *Fusobacterium*, (j) *Granuicatella*.

comparison with oral cancer microbiome has been scarcely studied. We, therefore, hypothesized that the oral microbiome of tobacco chewers must act as an intermediate stage between control and oral cancer microbiome.

In this pilot study, we have provided a comprehensive profile of oral microbiome of patients having oral cancer and long-term tobacco chewers as compared to the control group of an Indian population. The predominant phyla observed were *Firmicutes*, *Proteobacteria*, *Bacteroidetes*, *Fusobacteria*, and *Actinobacteria*. Phyla *Firmicutes* was most abundantly observed in the oral microbiome of control population, whereas it was third most abundant phyla in tobacco and OC group, which differed from previously published data for other studies where *Firmicutes* makes up maximum microflora for both healthy as well as OC patients [21, 22]. The most abundant flora found in tobacco chewers and OC patients was *Proteobacteria* followed by phyla *Bacteroidetes*, and for OC group was phyla *Bacteroidetes* followed by phyla *Proteobacteria*, which is not in concordance with data published before. This depicts the diversity and variation in the Indian oral microbiome, which could be attributed to many factors including lifestyle, food habits, etc. At the genus level, the abundance of *Streptococcus* decreased in OC and tobacco as compared to control. Genus *Streptococcus* accounts for a major population of commensal bacteria in the oral microbiome, which in this case has clearly declined to depict dysbiosis and possible cause for a rise in pathogenic bacteria in the non-control group [30]. Although oral *Streptococci* is not novel to the Indian population and is a common oral bacterium found in various populations across the globe, the high abundance in the healthy population could be attributed to the carbohydrate-rich diet of the Indian population. Sugars are metabolized to acids by supragingival saccharolytic bacteria, including *Streptococcus*. The contribution of carbohydrates from the host diet to microbial growth in the oral cavity is not well understood, but there is evidence that carbohydrate is a growth-limiting substrate for oral *Streptococci* [31–33]. Another genus *Neisseria* was found to have increased abundance in OC and tobacco groups as compared to the control group, which has been reported previously [22, 30]. Studies



Groups	Number of genera	Genera
Control-OC-Tobacco	44	*Streptococcus, Prevotella, Neisseria, Rothia, Haemophilus, Porphyromonas, g__1, Veillonella, Fusobacterium, Leptotrichia, Granulicatella, Actinomyces, Capnocytophaga, Actinobacillus, g__2, g__3, Aggregatibacter, Paludibacter, Lautropia, Campylobacter, Tannerella, Lactobacillus, Gemella, Treponema, g__6, Dialister, Atopobium, Oribacterium, Parvimonas, g__8, Corynebacterium, g__9, Filifactor, Peptostreptococcus, Megasphaera, Bulleidia, g__10, Selenomonas, Halomonas, Eikenella, g__12, g__13, Scardovia, Moryella*
Control-Tobacco	5	*g__5, g__7, Abiotrophia, g__11, g__14*
Tobacco-OC	3	*Catonella, Mycoplasma, Lachnoanaerobaculum*
OC-Control	2	*Bifidobacterium, Shuttleworthia*
Control only	1	*g__4*
Tobacco only	3	*g__16, g__17, g__18*
OC only	8	*Pseudomonas, Propionibacterium, g__15, Bacteroides, Morganella, Alloscardovia, Peptococcus, Klebsiella*

Genera labelled g__1 to g__18 are unidentified bacterial genus

Fig. 4. Venn diagram depicting common and unique number of bacteria.

have also reported the involvement of *Neisseria* species in alcohol metabolism, a known risk factor of oral cancer as it produces acetaldehyde, a known carcinogen [34]. Our results also depict the increase in abundance of genus *Porphyromonas* in OC and tobacco groups as compared to healthy individuals. *Porphyromonas gingivalis* is a well-studied and known periodontal pathogen [35, 36]. *P. gingivalis* attaches to the epithelial cells of the oral cavity, causes inflammation, evades host defence mechanisms by utilizing numerous virulence factors, and causes deregulation of innate immunity and inflammatory responses [37]. *P. gingivalis* along with *Treponema denticola* and *Tannerella forsythia* is a part of a red-complex which causes periodontitis [38].

Genus *Prevotella* was present in an increased proportion in OC and tobacco samples as compared to the control group. A few of the known species belonging to genus *Prevotella* that are associated with dental diseases and oral cancer include *P. melaninogenica* and *P.veroralis,* which have been reported to be potential biomarkers of the pre-cancerous stage [22]. *P. intermedia, P. nigresans, P. denticola* and *P. histicola* are few other oral pathogens associated with subgingival plaque, chronic periodontitis, alcohol metabolism etc. [39–41]. Our study also showed increased proportion of genus *Aggregatibacter* in OC and tobacco compared to control. This bacterium induces bone resorption using virulence factors such as Leukotoxin, LPS and enzymes in periodontal diseases [42]. The differential abundance of specific bacteria between the three groups depicts a transitional shift in the microbiome of the control to the OC via the tobacco chewing stage.

Genus *Rothia* showed a change in its abundance within the three study groups, with 10.06% in control and 6.96% in tobacco chewers whereas only 2.20% in the OC group. These results obtained are supported by a few previous studies where a significant reduction in the abundance of *Rothia* was observed in the cancer population as compared to healthy individuals [8, 30, 43]. Although a few other reports suggest the opposite, i.e. strong association with OC samples [44–46]. Therefore, the involvement of genus *Rothia* in oral cancer remains debateable.

Along with the well-known oral bacteria, a few uncommon bacterial genera were also identified in this study which showed differential abundances in their proportion in the three study groups. The presence of genus *Catonella* was observed in this study in the tobacco group and OC group. Although the abundance of this genus is low, it has been previously reported in a few studies. *Catonella morbi* is a Gram-negative bacillus which resides in the oral cavity as a commensal, but can act as an opportunistic pathogen and has a role in endodontic infections [47]. The increased abundance of this bacteria in OSCC cases as compared to healthy individuals has been previously reported [22, 48]. Another such genus is *Lachnoanaerobaculum*, which was reported in the present study in lower abundance but has been identified as

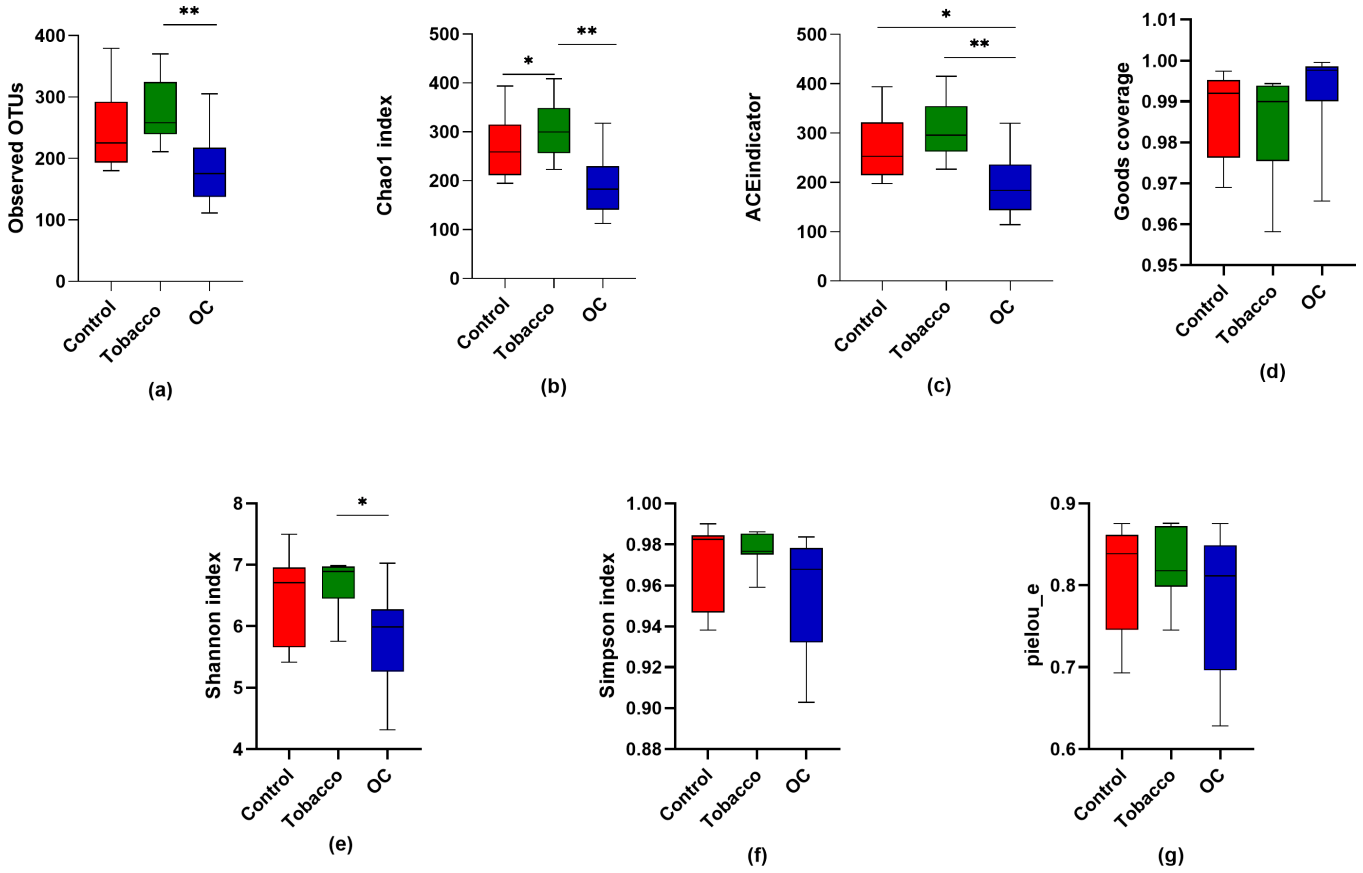


Fig. 5. Alpha diversity indices for the study groups; (a) Observed OTUs; (b) Chao1 index; (c) ACE indicator; (d) Goods coverage; (e) Shannon index; (f) Simpson index; (g) pielou_e.

a part of the OSCC microbiome in reported studies. Multiple species belonging to this genus have been associated with the oral microbiome such as *L. umeaense, L. saburreum, L. orale* and *L.* sp. *OBRC5-5* [22, 49, 50]. The genus has also been reported as a part of the core Indian salivary oral microbiome [51]. In this study, we observed the presence of the genus *Lachnoanaerobaculum* in tobacco and OC groups only.

The results obtained in this study are in parallel to most of the oral cancer metagenomic studies published previously. This signifies that the Indian oral cancer microbiome is similar to that in other parts of the world. Although, the additional study group i.e. the tobacco-chewing group shows an interesting result. The oral microbiome of long-term tobacco chewers was found to be similar to both the oral cancer microbiome as well the healthy microbiome. This pattern hints towards a transition of the oral microbiome from a healthy to an OC group, through the tobacco-chewing group, where the tobacco group shows intermediate (in-between) microbial composition, when comparing the alpha diversity indices and the bacterial composition. On the other hand, on observing the beta diversity plots, the diversity of tobacco chewers displays a similarity to control group. Hence no final judgement

Table 1. Comparison of alpha diversity matrices

Study group		Observed OTUs	ACE index	Chao1	Goods coverage	Shannon	Simpson	Pielou_e
Control	Mean	247.666	269.828	267.245	0.9872	6.43000	0.9699	0.81140
	SD	65.34	64.9824	64.9424	0.01060	0.71710	0.0205	0.06581
Tobacco	Mean	277.222	307.595	304.457	0.98384	6.66871	0.97777	0.82422
	SD	55.59	62.2552	61.2431	0.01267	0.41120	0.00841	0.04337
OC	Mean	180.777	191.744	188.896	0.9925	5.81041	0.95670	0.78061
	SD	61.49	64.6472	64.5031	0.01075	0.82789	0.03106	0.09107

Table 2. Comparison of beta diversity based on unifrac distances

Comparison groups			Permanova		Anosim		Permdisp	
			pseudo-F	*P*-value	R	*P*-value	F-value	*P*-value
Control	OC	Unweighted	1.7941	0.017	0.180384	0.011	9.105808	0.016
		Weighted	2.0164	0.093	0.21845	0.019	1.88538	0.123
Control	Tobacco	Unweighted	2.071999	0.009	0.216393	0.018	0.865104	0.339
		Weighted	1.535968	0.174	0.044925	0.213	0.2354	0.511
OC	Tobacco	Unweighted	2.378046	0.003	0.211591	0.008	15.12506	0.012
		Weighted	0.750181	0.534	0.038752	0.209	1.272984	0.256

can be made about the role of oral microbiome in tobacco chewers and their involvement in oral cancer development. Although the role of tobacco chewing in oral cancer is already a well-known phenomenon, exploring and investigating the microbial composition of tobacco chewers is necessary to support the role of microbiota along with tobacco-associated carcinogens towards the onset of oral cancer.

The demographic location, eating habits, and other lifestyle changes have been shown to play an important role in the

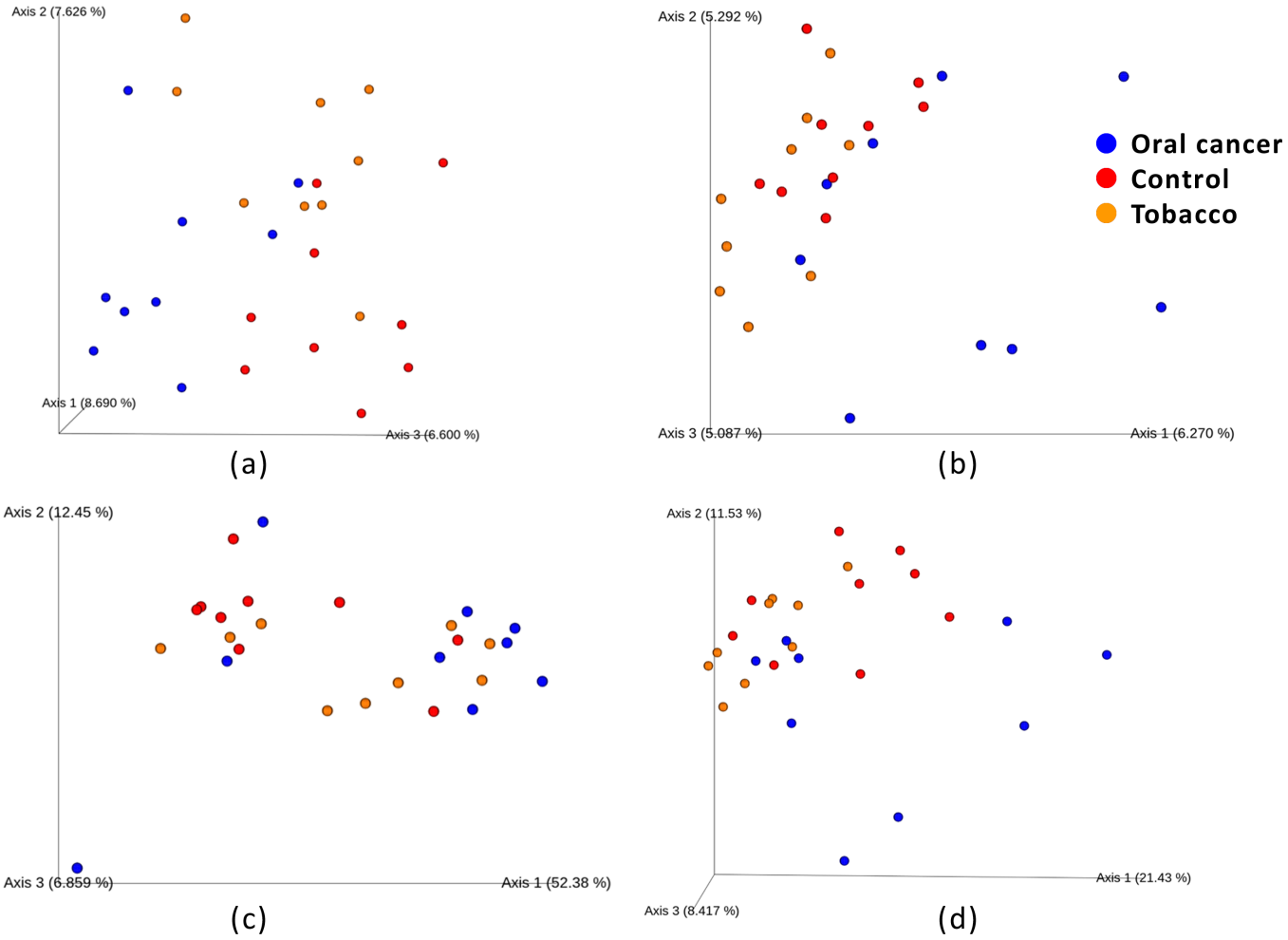


Fig. 6. Beta diversity PCoA plots depicting sample diversity between groups; (a) Bray-Curtis plot; (b) Jaccard matrix; (c) Weighted Unifrac matrix; (d) Unweighted Unifrac matrix.

dynamics of the oral microbiome [52]. The use of smokeless tobacco and areca nut, a regular habit in the United States and South Asia has been reported to play a role in oral cancer development. The tobacco-specific nitrosamines, namely N0-nitroso-nornicotine, 4-(Methylnitrosamino)−1-(3-pyridyl)−1-butanone) and the metabolite 4-(methylnitrosamino)−1-(3-pyridyl)−1-butanol are the strongest carcinogens in smokeless tobacco products and are labelled 'carcinogenic to humans' [53]. In India and Southeast Asia, tobacco is consumed in various forms such as paan, mawa, gutka, each of them possessing a varied risk of progression towards oral cancer. Gutka, which is sweet in taste is often used as a mouth freshener, but the sweetness leads to aggregation of various bacteria in the mouth leading to tooth damage. Similarly, consumption of tobacco in the form of paan which contains tobacco seeds, quenched lime, spices and areca nut folded in a betel leaf leads to pathophysiological and cellular changes in the oral cavity. Areca nut causes ROS production and hampers collagen synthesis and degradation pathways leading to increased collagen synthesis and inhibition of collagen degradation thereby causing oral submucous fibrosis, a precancerous disorder. Tobacco causes increased gingival blood flow and fibroblast damage whereas slaked lime causes cellular turnover, ROS production and chromosomal DNA damage [54, 55]. These ingredients not only cause host cell damage but also lead to dysbiosis in the oral microbiota. The abundance of oral bacteria like *Eubacterium nodatum, Peptostreptococcus micros, Streptococcus anginosus, Streptococcus constellatus, Gemella morbillorum, S. sanguinis, Veillonella parvula, P. micros, F. nucleatum, Fusobacterium periodonticum, and G. morbilorum, Treponema socranskii* and *Porphyromonas gingivalis* has been observed to increase after their interaction with tobacco extracts [14]. These bacteria are either known periodontal pathogens or have shown an increased abundance in oral cavity of oral cancer cases as well.

Recent research developments have pointed out that indeed, the microbiome population has the potential to affect or alter the human health and immune response and thus induce diseases and certain cancers [56–59]. However, the precise role of specific species and the mode/pathway of action of the microbiome leading to the development of cancer is still not clearly deciphered and understood. The reported data articulates that there exists diversity between cancer and non-cancer population from different community individuals. Beyond this, however, further understanding is required to gain strong insights into the functional characterization and mechanisms involved in the establishment of different microbial communities, understand the interspecies interaction and microbe-host interactions, which would eventually help in developing preventive or treatment therapeutics. Typically, in cancer patients, the oral flora diversity and load is affected by various factors like the disease and malignancy stage, antibiotic use, surgery or chemo-radiation treatment, age and overall health status. There is a dearth of reports that give information about microbiome analysis and its effect on oral cancer progression, treatment or morbidity.

Conclusion

Current studies on oral microbiome can help in developing microbe screening tests using simple saliva sample swabs from patients. This can be an attractive option as oral samples are easier to take, are usually non-invasive and is an inexpensive way of collecting samples as opposed to invasive methods. Even though immense research is being conducted worldwide to understand oral cancer and its nuances, there is still an enormous need to fully understand the aetiology and the role of microbes, understand the interspecies interactions and host-microbe interaction which could eventually help in diagnostics, prevention, and treatment therapies. The microbiome analysis of oral cancer patients and its correlation with premalignant conditions and oral cancer is anticipated to be useful in predictive diagnostics, progression and therapeutic regimen, or identification of therapeutic targets in the precancerous stages and oral cancers. This database for the Indian population generated by the study can help in developing a microbiome-based treatment strategy in oral cavity related complications and diseases.

Since this is a pilot study, limited information could be deciphered from current work. Although the results obtained in this study are in concordance with the previously published reports, along with an insight about the microbiota of tobacco chewers, future studies are required with more number of samples for better confidence in the results.

Data availability

The 16S rRNA amplicon sequencing data from this study have been deposited in the NCBI BioProject under accession number PRJNA751046.

Funding information
This research work was supported by research grant from NMIMS University seed grant (University Seed Grant 2016) and Seoul Women's University (2021).

Acknowledgements
The authors would like to thank Dr Suresh Shettigar, BARC Hospital, Mumbai for providing initial guidance with NGS analysis. The authors would also like to thank the working staff of Somaiya Ayurvihar-Asian Cancer Institute (K. J. Somaiya Medical College and Hospital, Mumbai) for providing help with oral cancer sample collection.

Conflicts of interest
The authors declare that there are no conflicts of interest.

Ethical statement
Informed written consent was obtained prior to the sample collection from all the participants. Ethical approvals for the recruitment of healthy, long-term tobacco chewing volunteers and oral cancer patients were granted by SVKM's Institutional Ethics Committee (NMIMS/IEC/008/2016, dt. 4/7/2016) and K. J. Somaiya Medical College and Hospital, Mumbai Institutional Ethics Committee (dt. 14/9/2017).

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